

Lecture 5

"Writing An Engineering Report"

Presented by

Dr. Kareem Kamal

WRITING AN ENGINEERING REPORT

In this lecture, we discuss **guidelines** for writing the standard components of an Engineering report:

1. **Transmittal letter**
2. **Covers and labels**
3. **Title page**
4. **Page Numbering**
5. **Executive summary**
6. **Table of contents**

7. **List of Figures and Tables**

8. **Introduction**

9. **Body of the report**

10. **Conclusions**

11. **References**

12. **Appendixes**

I. LETTER OF TRANSMITTAL

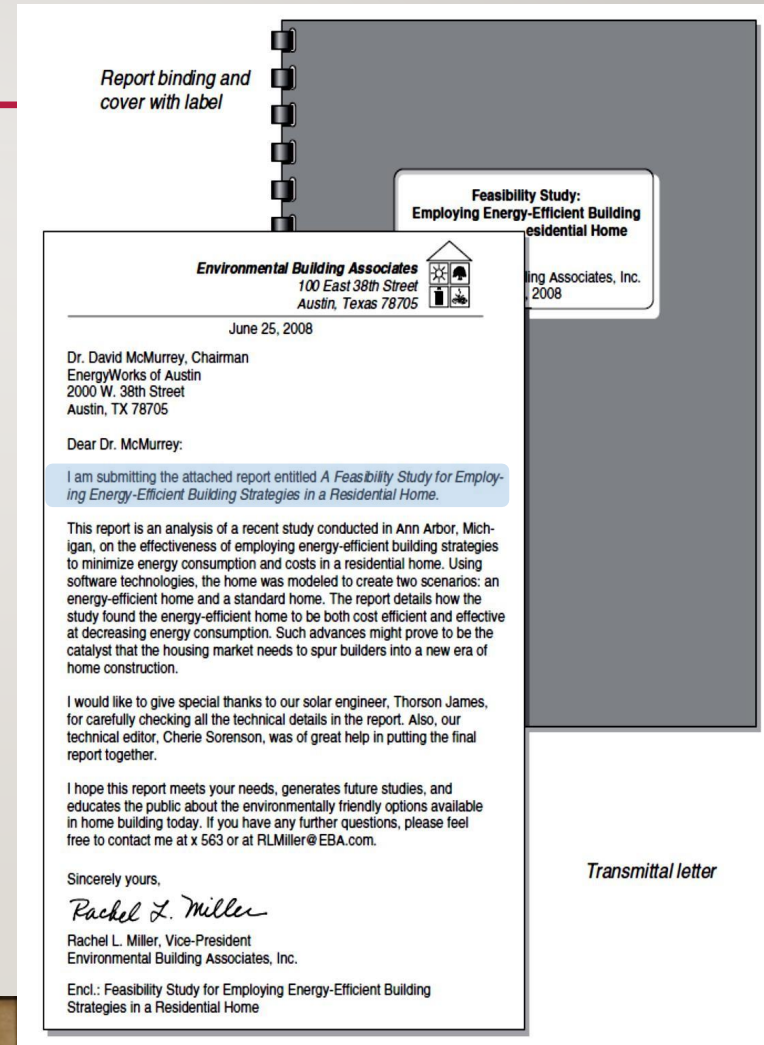
- The transmittal letter is a cover letter.
- It is either attached to the outside of the report or bound within the report.
- It is from the report writer to the recipient.

I. LETTER OF TRANSMITTAL

- The contents and the organization of the transmittal letter are as follows:

1. Paragraph 1.

- State the name of the report in *italics*.
- Mention the report-assigned date and the reasons for the report.



I. LETTER OF TRANSMITTAL

2. Paragraph 2

Focus on the purpose of the report and briefly describe the report contents.

3. Paragraph 3

Acknowledge any funding or help by other people, and mention any limitations or omissions in the report.

Report binding and cover with label

Feasibility Study:
Employing Energy-Efficient Building
Strategies in a Residential Home

Environmental Building Associates
100 East 38th Street
Austin, Texas 78705



Environmental Building Associates, Inc.
2008

June 25, 2008

Dr. David McMurrey, Chairman
EnergyWorks of Austin
2000 W. 38th Street
Austin, TX 78705

Dear Dr. McMurrey:

I am submitting the attached report entitled *A Feasibility Study for Employing Energy-Efficient Building Strategies in a Residential Home*.

This report is an analysis of a recent study conducted in Ann Arbor, Michigan, on the effectiveness of employing energy-efficient building strategies to minimize energy consumption and costs in a residential home. Using software technologies, the home was modeled to create two scenarios: an energy-efficient home and a standard home. The report details how the study found the energy-efficient home to be both cost efficient and effective at decreasing energy consumption. Such advances might prove to be the catalyst that the housing market needs to spur builders into a new era of home construction.

I would like to give special thanks to our solar engineer, Thorson James, for carefully checking all the technical details in the report. Also, our technical editor, Cherie Sorenson, was of great help in putting the final report together.

I hope this report meets your needs, generates future studies, and educates the public about the environmentally friendly options available in home building today. If you have any further questions, please feel free to contact me at x 563 or at RLMiller@EBA.com.

Sincerely yours,

Rachel L. Miller

Rachel L. Miller, Vice-President
Environmental Building Associates, Inc.

Encl.: Feasibility Study for Employing Energy-Efficient Building Strategies in a Residential Home

Transmittal letter

I. LETTER OF TRANSMITTAL

4. Last Paragraph

Express hope that the reader finds the report satisfactory.

Encourage the reader to get in touch with any questions, comments, or concerns.

Give your contact information.

Report binding and cover with label

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Transmittal letter

2. COVER & LABEL

- Covers

Use covers that allow reports to lie open by themselves.

This type uses a plastic spiral for the binding and thick card-stock paper for the covers

Don't forget to add a label for the cover of your report.

Report binding and cover with label

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Employing Energy-Efficient Building
Strategies in a Residential Home

Environmental Building Associates
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Austin, Texas 78705



Environmental Building Associates, Inc.
2008

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Encl.: Feasibility Study for Employing Energy-Efficient Building Strategies in a Residential Home

Transmittal letter

2. COVER & LABEL

- Labels.

Print it out, then photocopy it directly onto your report cover.

Your label should contain the report title, your name, your organization's name, and a date.

Report binding and cover with label

<i>Report binding and cover with label</i>	
Environmental Building Associates 100 East 38th Street Austin, Texas 78705 June 25, 2008 Dr. David McMurrey, Chairman EnergyWorks of Austin 2000 W. 38th Street Austin, TX 78705 Dear Dr. McMurrey: I am submitting the attached report entitled <i>A Feasibility Study for Employing Energy-Efficient Building Strategies in a Residential Home</i>. This report is an analysis of a recent study conducted in Ann Arbor, Michigan, on the effectiveness of employing energy-efficient building strategies to minimize energy consumption and costs in a residential home. Using software technologies, the home was modeled to create two scenarios: an energy-efficient home and a standard home. The report details how the study found the energy-efficient home to be both cost efficient and effective at decreasing energy consumption. Such advances might prove to be the catalyst that the housing market needs to spur builders into a new era of home construction. I would like to give special thanks to our solar engineer, Thorson James, for carefully checking all the technical details in the report. Also, our technical editor, Cherie Sorenson, was of great help in putting the final report together. I hope this report meets your needs, generates future studies, and educates the public about the environmentally friendly options available in home building today. If you have any further questions, please feel free to contact me at x 563 or at RLMiller@EBA.com. Sincerely yours, <i>Rachel L. Miller</i> Rachel L. Miller, Vice-President Environmental Building Associates, Inc. Encl.: Feasibility Study for Employing Energy-Efficient Building Strategies in a Residential Home	Feasibility Study: Employing Energy-Efficient Building Strategies in a Residential Home Environmental Building Associates, Inc. June 25, 2008 <i>Transmittal letter</i>

3.TITLE PAGE

- It's the first page on your report just after the front cover.
- Like labels, your title page should contain the report title, your name, your organization's name and logo (if it exists), and a date.



EarthLink™

EarthLink Interactive Voice Response System

1-888-EarthLink

Final Usability Report

11 December 2006

Submitted to

Dr. Marj Davis, Chair
Technical Communication Department
Mercer University

Submitted by

Katie Derthick, Team Leader
Karen Crow
Jason Daniel
Mercer University

4. PAGE NUMBERING

-
- All pages in the report, other than the covers, are numbered; although, numbers are not displayed on some pages.
 - Don't display numbers on the title page and page one of the introduction.
 - Use the traditional design, where all pages are numbered in Arabic numerals, except for the pages before the introduction (first page of the report body); they are numbered in lowercase Roman numerals.
 - It is preferred to place the page number at the bottom center of the page.

5.ABSTRACT & EXECUTIVE SUMMARY

- Abstracts summarize the report contents
- A report might have one or two abstracts, in which case each abstract plays a different purpose.

EXECUTIVE SUMMARY

This feasibility report analyzes a recent study conducted on a 2,450 ft² residential home (referred to as SH or standard home) built in Ann Arbor, Michigan. The goal of the study was to determine the effectiveness of employing energy-efficient building strategies to minimize energy consumption and costs in a residential home.

The home was modeled using Energy-10, a sophisticated software package capable of calculating the energy consumed during the use of the home over a 50-year period. While keeping the basic functional units (such as floor plan, occupancy, type and number of appliances, and internal volume) of the home consistent, SH was then modeled to reduce the energy consumption by employing various energy-efficient strategies (referred to as EEH or energy efficient home).

The total life-cycle energy consumption of SH was found to be 15,455 GJ, which consisted of space and water heating and cooling, lighting, ventilation, and appliances. The total life-cycle energy consumption of EEH was reduced to 5653 GJ. The purchase price of SH was \$240,000 (actual market value) and was determined to be \$22,801 more for EEH. The cost analysis performed found that despite a 9.5% increase in the purchase price of an energy-efficient home, lower annual energy expenditures make the present value nearly equal to the more energy-consuming version. The accumulated life cycle costs are higher in EEH until year 48 and are \$1,054 (or 0.1%) less at year 50.

It was found that the most effective strategy for reducing overall annual energy costs is installation of a high-efficiency HVAC system. However, for reducing overall energy consumption, insulation was the most effective strategy followed by high-efficiency HVAC and air leakage control.

5.ABSTRACT & EXECUTIVE SUMMARY

Abstracts are

- **Descriptive abstract**
Provides an overview of the purpose and the report contents.
- **Executive summary** Summarizes the key facts and conclusions in the report.

The executive summary should not go over three pages.

EXECUTIVE SUMMARY

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6. TABLE OF CONTENTS

- The Table of Contents (*TOC*) shows readers:

- What **topics** are covered in the report?
- How those topics are discussed (the **subtopics**).
- On which **page numbers** do those sections start.

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EXECUTIVE SUMMARY.....	ii ←
LIST OF FIGURES AND TABLES.....	iv
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2.3.2 Energy-efficient strategies.....	
2.4 Energy Consumption Determination.....	
2.4.1 Heating and cooling systems.....	
2.4.2 Electrical systems.....	
3.0 CONSUMPTION COMPARISONS.....	
3.1 Gas Consumption.....	
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4.0 COST ANALYSIS.....	
4.1 Determination of Cost.....	
4.1.1 Construction.....	
4.1.2 Energy costs.....	
4.2 Accumulated Cost Analysis.....	
5.0 RANKING OF ENERGY-EFFICIENT HOMES.....	
6.0 CONCLUSIONS.....	
REFERENCES.....	

EXECUTIVE SUMMARY	
This feasibility report analyzes a residential home (referred to as Standard Home) in Ann Arbor, Michigan. The goal of the report is to assess the feasibility of employing energy-efficient systems to reduce energy consumption and costs in the home.	
The home was modeled using EnergyPlus, a software package capable of calculating the energy consumption of the home over a 50-year period. Input data (such as floor plan, occupancy, and internal volume) of the home were obtained from the homeowner. The report identifies strategies to reduce the energy consumption and costs of the home (referred to as Energy-Efficient Home).	
The total life-cycle energy consumption of the Standard Home is 1,000,000 GJ, which consisted of space heating, ventilation, and appliances. The total energy consumption of EEH was reduced to 5653 GJ. The total cost of EEH was \$240,000 (actual market value) and the cost savings for EEH. The cost analysis predicts a 40% increase in the purchase price of a home with EEH.	

6. TABLE OF CONTENTS

- Don't crowd the TOC.
- Don't include more than three levels of headings in the TOC.
- Make sure the words in the TOC are the same as they are in the text.
- Each of the three levels of headings are aligned together.
- Page numbers are right-aligned with each other.

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The home was modeled using EnergyPlus, a software package capable of calculating the energy consumption of the home over a 50-year period (such as floor plan, occupancy, and internal volume) of the home to reduce the energy consumption of the home (referred to as EEH).

The total life-cycle energy consumption of the EEH was 5653 GJ, which consisted of space and ventilation, and appliances. The total cost of EEH was reduced to \$240,000 (actual market value) and more for EEH. The cost analysis shows that the increase in the purchase price of the EEH is offset by the reduction in the purchase price of the EEH.

6. TABLE OF CONTENTS

- First-level sections are all caps
- Second-level headings use initial caps on each main word.
- Lower-level sections use initial caps on the first word only.
- First-level sections have extra space above and below.

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7. LIST OF FIGURES & TABLES

- It helps the reader to easily find figures and tables within your report.
- Create separate lists of figures and tables, for long reports.
- For short reports, put them on the same page if they fit and combine the two lists under the heading, “List of Figures and Tables.”

LIST OF FIGURES

Figure 1. Natural Gas Use by SH and EEH.....	7
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Table 2. Energy-10 Simulation.....	
Table 3. Energy-Efficient Strategies.....	
Table 4. Cost Comparisons for SH and EEH.....	

1.0 INTRODUCTION

1.1 Purpose of the Report

This report analyzes the results of the simulation strategies to determine if such strategies can reduce the amount of energy consumed in the amount of energy consumed. It analyzes which home system strategies result in the greatest reductions in energy and whether they are efficient in the long run.

1.2 Background of the Report

Annually, 24% of the natural gas is consumed by the residential sector. This is equivalent to 1.5 metric tons of greenhouse gas per person. Understanding energy consumption and its environmental impacts is essential if a systematic approach to reducing energy consumption will not only reduce environmental impact on the environment.

8. INTRODUCTION

- The introduction is an essential part of any report; it prepares the reader to read the main body of the report.
- The introduction discusses the following:
 1. Specific purpose and topic of the report.

RES

EH..... 7

and EEH..... 8

Topic overview: Always provide a brief idea of the contents of the report in the introduction.

1.0 INTRODUCTION

1.1 Purpose of the Report

This report analyzes the results of using various energy-efficient strategies to determine if such practices actually make a difference in the amount of energy consumed by an average house. Additionally, it analyzes which home system improvements provide the greatest reductions in energy and whether such improvements are cost-efficient in the long run.

1.2 Background of the Report

Annually, 24% of the natural gas and 35% of the electricity in the US is consumed by the residential housing sector. Consequently, 1.3 metric tons of greenhouse gases are emitted annually [6, 7]. Understanding energy consumption and taking measures to reduce it is essential if a systematic and comprehensive reduction of environmental impacts is desired. Reductions in home energy consumption will not only reduce utility costs but also reduce the impact on the environment.

1.3 Scope of the Report

This report provides technical background on the construction of the standard and the energy-efficient house, the energy-efficient strategies used in the latter, energy-consumption rates, construction costs, and other relevant details. Not included in this report are discussions of the receptiveness of the American home-building industry or American home buyers to energy-efficient housing design or of pending legislation to promote energy-efficient housing design.

Note: A basic understanding of terminology for housing constructing, HVAC, and cost analyses is assumed.

8. INTRODUCTION

2. **Intended audience** of the report (*Knowledge or experience that readers need in order to understand the report*).
3. **Motivation** of the report.
4. **Scope** of the report (*topics included*).
5. **Background** (*concepts, definitions, history, statistics*).

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8. INTRODUCTION

- The introduction should not be more than 2 pages for a 20 page report.
- The background should be 1/3 of the introduction.
- If the background needed is large, move it to a section of its own.

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9. BODY OF THE REPORT

- The body of the report is the **main text of the report**, the sections between the introduction and conclusion.
- Make sure that the report body contains
 1. Heading
 2. Lists
 3. Symbols, numbers, and abbreviations
 4. Sources of borrowed information
 5. Graphics and figures
 6. Cross-references

10. CONCLUSIONS

- This is the final section in your report. It should combine any of the following functions:
-

➤ Conclude

Draw **logical conclusions from the data** and the discussion previously presented within the report.

➤ Summarize

Review the key points from the previous material.
Summaries present nothing new.

10. CONCLUSIONS

➤ Generalize

Move away from the specific topic of the report to a **general discussion** of implications, applications, and future development.

- The **length** of the conclusion section can be **from a 100-word paragraph up to five or six pages**.
- **One or two pages** are enough for a **10 to 20-page** report.

II. REFERENCES PAGE

- In this section, your information sources are listed in a numerical order according to when they are first cited in the text.

energy consumption of

Source of the borrowed information at the bottom of the conclusion page.

REFERENCES

- [1] S. Blanchard and P. Reppe, "Life cycle analysis of a residential home in Michigan." Master's thesis originally published October 1998. <http://www.umich.edu/~nppcpub/research/lcahome>. Accessed July 4, 2003.
- [2] Energy-10, Release 1.2, Software application. Passive Solar Industries Council, 1511 K Street. NW, Suite 600, Washington, DC 20005. January 1998.
- [3] National Association of Home Builders, *Building Greener, Building Better: The Quiet Revolution*. http://www.nahb.org/assets/docs/publication/greener_232003121230PM.pdf. Accessed July 4, 2003.
- [4] National Association of Home Builders, "Housing facts and figures: characteristics of new single family homes, 1971–1994." <http://nahb.com/sf/html>. Accessed May 21, 1998.
- [5] J. D. Nisson, *The Super-Insulated Home Book*. New York: Wiley, 1995.
- [6] P. Pierquet, J. Bowyer, P. Huelman, "Thermal performance and embodied energy of cold climate wall systems." *Forest Products Journal*, vol. 48, no. 6, pp. 53–60. June 1998.
- [7] U.S. Department of Energy, *Annual Energy Review: 1996*. "Table 6.6 Natural gas consumption by use sector, 1949–1997." <http://tonto.eia.doe.gov/FTPROOT/multifuel/038496.pdf>. Accessed July 3, 2003.
- [8] U.S. Department of Energy, *Annual Energy Review: 1996*. "Table 8.9 Electric utility retail sales of electricity by end-use sector, 1949–1997." <http://tonto.eia.doe.gov/FTPROOT/multifuel/038496.pdf>. Accessed July 3, 2003.

12.APPENDIXES

-
- Appendixes are those extra sections following the list of reference page.
 - Put in the appendixes anything that might distract and interrupt the flow of the report but cannot be left out of the report altogether.

I 2.APPENDIXES

-
- Appendixes are commonly used for
 1. Large tables of data
 2. Big chunks of sample code
 3. Fold-out maps
 4. Background that is too basic or too advanced for the body of the report
 5. Large illustrations that don't fit in the body of the report

Thanks for your Attention

Presented By: Dr. Kareem kamal